

Omkar Jahagirdar

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EDUCATION

Stony Brook University

Master of Science in Computer Science (Concentration in Data Science)

Stony Brook, NY

Aug. 2024 – May 2026

Savitribai Phule Pune University

Bachelor of Technology in Computer Science

Pune, India

Aug. 2020 – May 2024

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C/C++, SQL, Swift

Generative AI: LLMs (GPT-4o, LLaMA2-7B, Mistral-7B), LangChain, HuggingFace, PyTorch, TensorFlow, RAG, Fine-tuning

Frameworks: Django, Spring Boot, Pandas, Flask, ReactJS, SwiftUI, Node.js

Tools and Technologies: AWS, Google Cloud, Git, Docker, Kubernetes, Terraform, Jenkins, Spinnaker

Databases: PostgreSQL, MySQL, MongoDB, Firebase, Firestore

EXPERIENCE

Software Engineering Intern

May 2025 – Aug. 2025

SculptSoft LLC.

San Francisco, CA

- Developed an **AWS Kinesis**-based real-time ETL pipeline processing **50,000+** events per minute, integrating schema standardization, cleansing, and transformation logic; built a **rule engine** executing **80+** business rules to detect anomalies, violations, and generate detailed operational reports.

Software Engineering Intern

Jan. 2024 – Jun. 2024

JPMorgan Chase & Co.

Mumbai, India

- Designed and implemented ETL pipelines to process over **20GB** of data daily from **85+** sources, resulting in **\$150K+** annual cost savings by decommissioning legacy workflows.
- Engineered the backend architecture for a novel data correction tool employing **Spring Boot** and **Aurora PostgreSQL** on **AWS ECS**.
- Resolved security vulnerabilities using **Raven**, increased test coverage to **95+%**, and automated infrastructure with **Terraform** and **CI/CD** via **Jenkins** and **Spinnaker**, achieving a **50%** faster deployment cycle.

Software Engineering Intern

Jun. 2023 – Jul. 2023

JPMorgan Chase & Co.

Bangalore, India

- Developed an end-to-end system by fine-tuning a **BERT** model to identify data compliance violations, insider trading, and information leaks by employees.
- Achieved **92%** accuracy using **transfer learning** on unlabelled email and chat data.
- Developed a compliance dashboard using **Dash** to identify and visualize inter-employee violation patterns; adopted as a prototype for compliance regulations.

PROJECTS

Mobile Application Permission Prediction System | Python, Transformers, LLMs, LangChain

- Built an NLP-based multi-label permission prediction pipeline using **HuggingFace Transformers**, **DistilBERT**, **DeBERTa** on **5,400+** Android apps; evaluated **LLMs (GPT-4o, LLaMA2-7B, Mistral-7B)** and analyzed class imbalance and error patterns across permission categories.

Wrist Jockey: Control Music from Your Wrist | Swift, SwiftUI, Node.js, CoreMotion, Spotify API

- Architected a full-stack gesture-recognition system spanning **watchOS**, **iOS**, and **Node.js**, streaming calibrated **6-axis motion data** from Apple Watch to enable real-time, on-device gesture classification.
- Implemented a **Dynamic Time Warping** classifier with **Barycenter Averaging** and adaptive per-gesture thresholds for personalized, noise-robust recognition from few samples; integrated **Spotify OAuth 2.0** for real-time playback control.

Enterprise Innovation Workflow System | Python, Django, Tailwind CSS, AWS

- Architected a full-stack **idea-management platform** for **Bridgestone** across 4 user roles with automated email workflows, **granular RBAC**, and secure authentication for enterprise-grade multi-tenant workflow orchestration.